

Run your own Web and DB servers using XAMPP

Lecture 30

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Outline

- Review final exam answers
- Backup and restore
 - Run your own MySQL server or XAMPP
 - Run your projects on your laptop
- Copy your project and tables to your own PC
- Web Hosting + MySQL
- Download and install XAMPP
- Start web service
 - verify htdocs folder
 - Understand the configuration and services

Review final exam answers

- Average
- Stdev
- Final grade

Important notes about accounts

- Your Kean domain account will be inactive if you graduate or not taking any courses.
 - That means you cannot use VPN.
 - You won't be able to login to obi.kean.edu.
 - Your DB accounts might be available for a short time, but not too long.
 - Your project might be alive for a short time, but too long. You need to update URL on your resume.
- Therefore, you must backup today:
 - All PHP, HTML, any files from obi to your PC.
 - Tables from imc to your own mysql server on your PC.
- Instructors are not responsible to backup those data for you.

Web hosting service

- Example: Godaddy.com
 - Select Linux system
- Domain name
 - <https://www.namesilo.com/>
 - <https://www.godaddy.com/domains/domain-name-search>
- Web hosting
 - GitHub pages - <https://pages.github.com/>
 - <https://www.godaddy.com/hosting/web-hosting>
- Database server
 - <https://www.godaddy.com/help/create-mysql-or-ms-sql-databases-36>
- Price

Web Hosting + MySQL

- Web Hosting
 - Personal or business
 - Functions: Storage limit, PHP/Java support, MySQL, ...
 - Interface access the accounts: ssh or sftp only
 - Database management: user and database
 - Emails
 - Support
- Godaddy, Amazon, iPage,

XAMPP

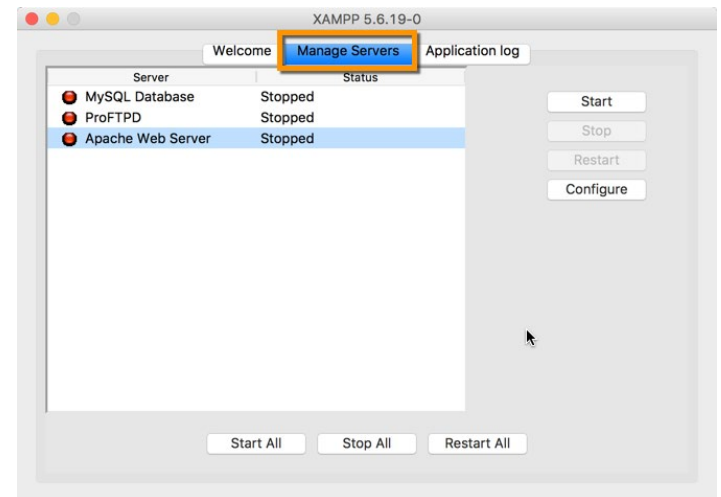
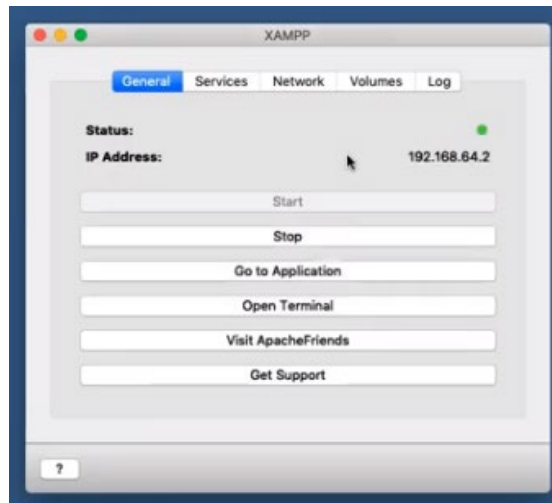
- XAMPP is free and open source cross platform web server solution stack package
 - X (to be read as "cross", meaning cross-platform)
 - Apache HTTP Server (web)
 - MySQL (database)
 - PHP (interpreters for scripts)
 - Perl (programming languages)
- Developer: Apache Friends
- Download at <https://www.apachefriends.org/index.html>

Install XAMPP on Windows

- How to Install XAMPP Server on Windows 10 [2021 Update] Step by Step Installation guide
 - <https://www.youtube.com/watch?v=TDiZWoiwk>

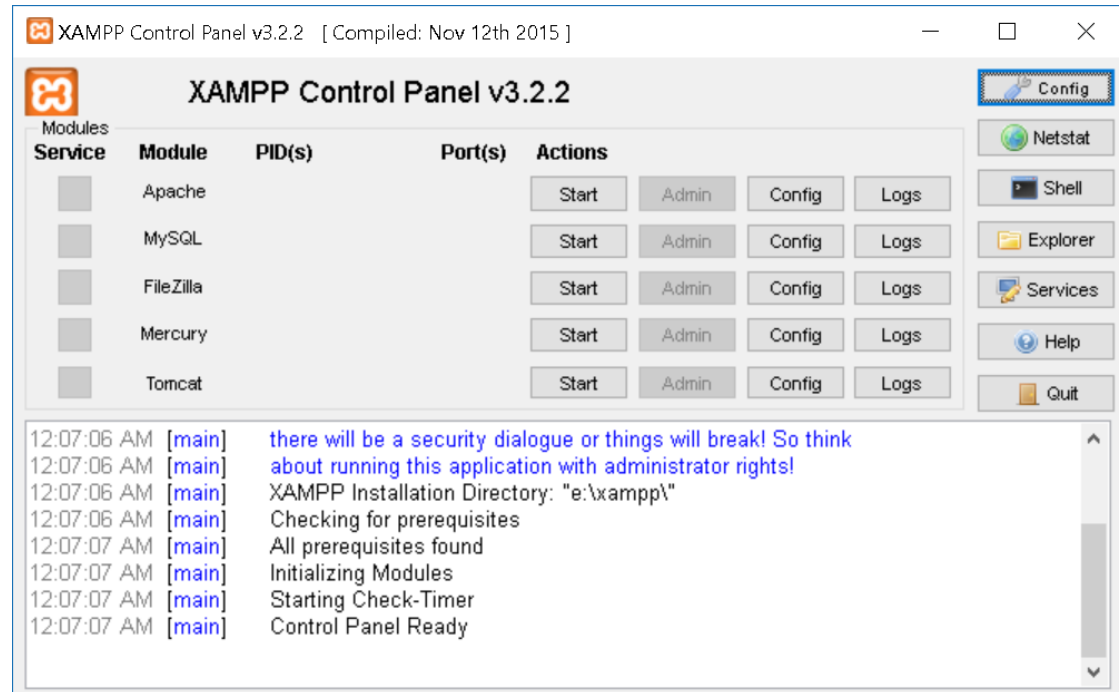
Install XAMPP on Mac

- How to install, start, and test xampp on Mac
 - <https://www.webucator.com/how-to/how-install-start-test-xampp-on-mac-osx.cfm>
- Some Mac requires to install mysql server
 - <https://dev.mysql.com/downloads/>
- YouTube: <https://www.youtube.com/watch?v=yyjBFiXsOkI>



Windows XAMPP Control Panel

- Download
- MySQL DB server
- Apache web server
- Shell
- Service – start, stop
- Configuration



- Standalone database, for personal test only!
- Project's database must be on the assigned server

MySQL server configuration

- Use XAMPP MySQL server to exam the configuration file:
 - The port number
 - Data folder
 - Log files

Apache web server configuration

- The port number
- What/Where is the Document Root?
- CGI-bin folder
 - For Python, NOT for PHP
- Log files – access and error
- Start and stop service

Your local website directory

- MS-Windows
 - C:\xampp\htdocs
- Mac - where to find/place local web site
 - /Applications/XAMPP/xamppfiles/htdocs
 - <https://www.tech-otaku.com/local-server/where-to-place-local-web-site-content-xampp-mac/>

Transfer your project and run it on your laptop

1. Create workspace on your laptop
2. Backup related tables from database on `imc.kean.edu`
3. Start database on your laptop
4. Restore tables to your laptop
5. Backup PHP and HTML files from `obi.kean.edu`
6. Start Apache web server
7. Run your projects on your laptop
8. Test your project

Copy your project to htdocs

- Use FileZilla to copy your entire project folder from the Linux server folder to htdocs folder.
- Start Apache (httpd) service
- Check the URL on your laptop, **for example, if your project folder is CPS3740**
 - <http://localhost/CPS3740/>
- Run your index.html and PHP program under CPS3740
 - **Change your dbconfig.php in your localhost (Do NOT change dbconfig.php on obi server)**
Eg: `<?php`
 `$host="localhost";`
 :
 `?>`
- Home exercises:
 - Make a syntax error in the php program and reload the program to see the error message
 - Debug your PHP programs and transfer them to obi.kean.edu if there is no error.

MySQL server configuration

- The port number
- Data folder
- Log files

Backup your tables in MySQL

- Use command (cmd) on linux server or your local laptop

```
$ mysqldump -u xxxx -p -h imc.kean.edu dreamhome > dreamhome.sql
```

Error message: (how to fix?)

mysqldump: Got error: 1044: Access denied for user 'demo'@'%' to database 'dreamhome' when using LOCK TABLES

<Answer> google “mysqldump Lock table”

Hint: --single-transaction

Exercise - backup and restore dreamhome database

- Use command (cmd) on linux server or your local laptop

```
$ mysqldump --single-transaction -u xxxx -p -h imc.kean.edu  
dreamhome > dreamhome.sql
```

```
$ mysql -u xxxx -p -h Your_PC Your_DB < dreamhome.sql
```

- Use PHPMyAdmin
- <https://help.dreamhost.com/hc/en-us/articles/214395768-phpMyAdmin-How-to-import-or-restore-a-database-or-table>

Transfer your project and run it on your laptop: 1/3

1. Create workspace on your laptop

-- make a working folder under
xampp\htdocs\CPS3740

2. Backup related tables from database on imc.kean.edu

```
# mysqldump --single-transaction -u demo -p -h imc.kean.edu  
database_name TableNames > Output_name.sql
```

Eg:

```
# mysqldump --single-transaction=true -u demo -p -h imc.kean.edu  
dreamhome > dreamhome.sql
```

3. Start database on your laptop

4. Restore tables to your laptop

```
# mysql -u xxxx -p -h localhost Your_DB < Output_name.sql
```

Transfer your project and run it on your laptop: 2/3

- Repeat the same steps to backup and restore other required tables in the project

Transfer your project and run it on your laptop: 3/3

5. Backup PHP and HTML files from obi.kean.edu

-- Transfer your project PHP and HTML files from obi.kean.edu to your work folder

-- update all your PHP and HTML files for the web server and database server (eg: dbconfig.php)

web server from obi.kean.edu to local web server

database server from imc.kean.edu to local database server.

6. Start Apache web server

7. Run your projects on your laptop

8. Test your project

Note: You can transfer your project to your local server for your practice only, but I will grade your project on our linux server and imc server.

Exercises – admin on XAMPP

- Login as root
- Create a user “dba_tester”
- Create 2nd user “tester”
- Reset their passwords using password function
 - Google the syntax
- Create a database “testdb”
- Assign all privileges on testdb to dba_tester@% with grant option

Exercises – DCL on XAMPP

- Logout root and login as dba_tester
- dba_tester assigns CREATE, SELECT, INSERT privileges to tester1
- Logout dba_tester and login as tester1
- Try to create a table, insert some records, and select the data
- Try to update the data and see what happens.

Making Backups with mysqldump

- The program mysqldump can back up **all kinds of tables**.
 - `$ mysqldump [options] db_name [tbl_name ...] > dump.sql`
- To backup particular tables t1 and t2
 - `$ mysqldump -u User1 -p -h host1 dbname t1 t2 > t1t2_backup.sql`
 - User1 must have SELECT privilege on the backup tables.
- Backup of entire database including accounts, tables views and stored routines
 - `$ mysqldump -u User1 -p -h host1 dbname > all_backup.sql`
- To load the backup file back into a new server:
 - `$ mysql -u User2 -p -h host2 new_dbname < backup-file.sql`

Exercises using XAMPP

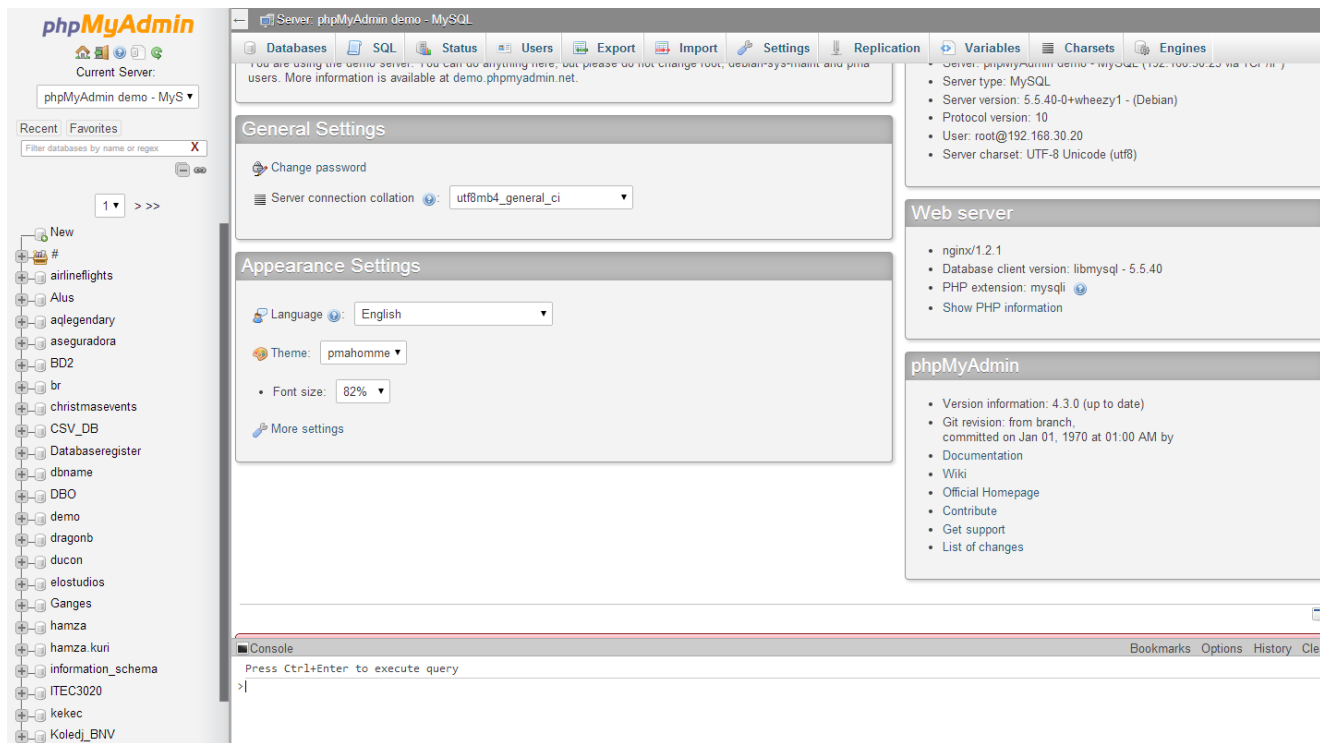
- Transaction exercises
- Dump dreamhome and your tables
- Backup your project
- Run your projects on your laptop
- Do exercises on your own MySQL server

MySQL Admin

- PHPMyAdmin
- <http://localhost/phpmyadmin/>
- Workbench connect to your local database server

Manage your own Database by PHPMyAdmin on Windows

- <http://localhost/phpmyadmin/>
- phpMyAdmin - Basics Tutorial
 - https://www.youtube.com/watch?v=1-81n_vuwug



Connect to remote server using PHPMyAdmin

- ***DO NOT TRY IT if you are not comfortable to change the settings

<http://www.devside.net/wamp-server/accessing-remote-databases-using-local-phpmyadmin>

- Download and use MySQL Workbench

Conclusion

- Backup and restore tables/databases to your own database server
- Copy your projects to your PC
- Understand MySQL Storage Engines